

LABORATORY OBSERVATION/RECORD

Course:	Full Stack Web Development	Course Code:	FSWD4121
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1. TITLE

Write a Program using JavaScript – Arrays and Functions (push, unshift, shift, spread, arrow functions)

2. AIM

To write a JavaScript program that demonstrates the use of arrays and functions, including array methods (push, unshift, shift, spread operator), regular functions, and arrow functions.

3. OBJECTIVE

The objectives of this experiment are:

- To declare and manipulate arrays using built-in JavaScript methods.
 - To use push() to add elements to the end of an array.
 - To use unshift() to add elements to the beginning of an array.
 - To use shift() to remove the first element of an array.
 - To use the spread operator (...) to create a new extended array.
 - To define and call regular functions with parameters.
 - To write and use arrow functions for concise expressions.
 - To display output using console.log().
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4. THEORY

JavaScript provides powerful built-in methods for array manipulation and supports multiple styles of defining functions.

- **Arrays:** Ordered collections that store multiple values. Indexed from 0.
 - **push():** Adds one or more elements to the end of an array.
 - **unshift():** Adds one or more elements to the beginning of an array.
 - **shift():** Removes and returns the first element of an array.
 - **Spread Operator (...):** Expands an array into individual elements, useful for copying or merging arrays.
 - **Regular Functions:** Defined with the function keyword; can accept parameters and return values.
 - **Arrow Functions:** A concise syntax using =>; ideal for simple one-line expressions.
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5. ALGORITHM / PROCEDURE

1. Create a new JavaScript file named record3.js.
2. Declare an array shoppingList with three string values.
3. Define a function addItem() that uses push() to add a new element.
4. Print the initial list, call addItem() twice, then print the updated list.
5. Declare a second array food with four string elements.
6. Use push() to add 'tandori' and unshift() to add 'biryani'.
7. Print the modified array.
8. Use shift() to remove the first element and print the result.

9. Use the spread operator to create an extended array and print it.
 10. Define a regular function greet(a, b) that adds two numbers (no return statement).
 11. Call greet() and log the result (should be undefined); also log the global result variable.
 12. Define an arrow function square = n => n * n.
 13. Call square(4) and log the output.
 14. Run the program using Node.js and verify all outputs.
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6. PROGRAM

record3.js

```
// Shopping list array and function
let shoppingList = ["Milk", "Bread", "Eggs"];
function addItem(newItem) {
  shoppingList.push(newItem);
}
console.log("Initial List:", shoppingList);
addItem("Bananas");
addItem("Cheese");
console.log("Updated List:", shoppingList);

// Food array and array methods
const food = ["pizza", "burger", "kfc", "frenchfries"];
console.log(food);
food.push("tandori");
food.unshift("biryani");
console.log("modified array is: ", food);
food.shift();
console.log(" array is: ", food);
const extend = [...food, "chicken curry", "pulav"];
console.log(extend);

// Regular function (no return statement)
function greet(a, b) {
  result = a + b;
}
console.log(greet(8, 9));
console.log(result);

// Arrow function
const square = n => n * n;
console.log(square(4));
```

7. CODE EXPLANATION

Code	Explanation
shoppingList.push()	Adds a new item to the end of the shoppingList array.
food.push('tandori')	Appends 'tandori' to the end of the food array.

<code>food.unshift('biryani')</code>	Inserts 'biryani' at the beginning of the food array.
<code>food.shift()</code>	Removes the first element ('biryani') from the food array.
<code>[...food, ...]</code>	Spread operator creates a new array by expanding food and adding more items.
<code>greet(a, b)</code>	Regular function that adds a and b but has no return statement, so it returns undefined.
<code>result = a + b</code>	Assigns sum to a global variable result (no let/const), accessible outside the function.
<code>console.log(greet(8,9))</code>	Prints undefined because greet() has no return statement.
<code>square = n => n * n</code>	Arrow function that takes n and returns its square. square(4) returns 16.

8. TEST CASES AND EXECUTION DATA

Test Case	Operation	Expected Output	Actual Output	Status
TC-01	Initial shoppingList	['Milk','Bread','Eggs']	['Milk','Bread','Eggs']	Pass
TC-02	After addItem x2	['Milk','Bread','Eggs','Bananas','Cheese']	['Milk','Bread','Eggs','Bananas','Cheese']	Pass
TC-03	push + unshift on food	6-element array with 'biryani' first	6-element array with 'biryani' first	Pass
TC-04	shift() removes first	5-element array, 'biryani' removed	5-element array, 'biryani' removed	Pass
TC-05	Spread to extend array	7-element array with curry & pulav	7-element array with curry & pulav	Pass
TC-06	greet(8,9) – no return	undefined	undefined	Pass
TC-07	Global result variable	17	17	Pass
TC-08	Arrow fn square(4)	16	16	Pass

9. OUTPUT

Output: Debug Console Execution

```

C:\Program Files\nodejs\node.exe --experimental-network-inspection .\record 3.js
> Initial List: (3) ['Milk', 'Bread', 'Eggs']
> Updated List: (5) ['Milk', 'Bread', 'Eggs', 'Bananas', 'Cheese']
> (4) ['pizza', 'burger', 'kfc', 'frenchfries']
> modified array is: (6) ['biryani', 'pizza', 'burger', 'kfc', 'frenchfries', 'tandori']
> array is: (5) ['pizza', 'burger', 'kfc', 'frenchfries', 'tandori']
> (7) ['pizza', 'burger', 'kfc', 'frenchfries', 'tandori', 'chicken curry', 'pulav']
undefined
17
16

```

10. RESULT

Thus, a JavaScript program was successfully written demonstrating the use of **arrays** and **functions**. Array methods including **push()**, **unshift()**, **shift()**, and the **spread operator** were demonstrated. Both regular functions and **arrow functions** were implemented and executed. The behavior of a function without a return statement (returning undefined) and the use of implicit global variables were also observed. All outputs matched the expected results.
